gravitational potential

Links closely with electric potential.

Gravitational potential at a point V_g is the work done W in moving unit mass from infinity to the point.

Gravitational potential difference ΔV_g between two points is given by $\Delta V_g = \frac{W}{m}$, where m is the mass and W is the work done in moving the mass between the points.

Unit of V_g is $\mathrm{J\ kg^{-1}}$. It is a **scalar** quantity.

$$V_g = \begin{cases} 0 & \text{at infinity} \\ < 0 & \text{otherwise} \because \text{grav. force is always attractive} \end{cases}$$

To move a test object **to** ∞ **from** somewhere **close to the mass producing** the field $\implies$ work must be done (to overcome the attractive force).

When a system has to be assembled from many components, the potential energy of the system is the work done to bring **every** component from ∞ to the final position.

W.r.t the formula book, the values for **gravitational potential** V_g and **gravitational potential energy** E_p apply to a radial gravitational field.

Close to a planet surface, $\exists$ a uniform gravitational field strength $\implies g = -\frac{\Delta V_g}{h}$, where h is the change in height of an object of mass m. Hence,

$$\begin{aligned} \Delta V_g &= -gh \\ m\Delta V_g &= -mgh \\ \therefore m\Delta V_g &= -\Delta E_p \qquad \text{(from topic A3)} \end{aligned}$$